

Feasibility Report – NatureScope

Introduction

NatureScope is an AI-powered handheld device designed for real-time species identification, primarily insects, but scalable to broader fauna. It integrates embedded AI vision, gamified learning (Pokémon card style), and an Ask-Me-Anything (AMA) mode using lightweight on-device LLMs.

Technical Feasibility

The system requires careful balance between performance and cost. The RK3576 SoC provides sufficient processing power for local inference, text-to-speech, and AMA mode. Dataset preparation involves cleaning the iNaturalist dataset (~4,700 species) and using YOLO-based detector-classifier pipelines. Inference target is ≤ 250 ms per frame.

Operational Feasibility

The device will be ergonomic, portable, and child-friendly. Housing will use ABS/Polycarbonate with 3D-printed prototypes. Snap-fit assembly ensures robustness. A 3000mAh Li-Po battery provides ~4–6 hours of field use.

Economic Feasibility

Estimated prototype cost: \$70–93 (including delivery). Target BOM: $\leq \$40$ for scalability. PCB manufacturing through JLCPCB and in-house 3D printing will reduce costs. Open-source software tools will be used.

Timeline & Milestones

- R&D; Phase: 640 hrs (component validation, dataset preparation, firmware integration).
- PCB Phase: 480 hrs (schematic, layout, fabrication, validation).
- Mechanical Design: 200 hrs (CAD, prototyping, validation).

Total Project Duration: ~4–6 months.

Risk Assessment

- Dataset gaps may reduce accuracy (mitigated by augmentation).
- RK3576 may struggle with AR; fallback to RK3588.
- Battery optimization needed for long use.
- Scaling challenges in sourcing and licensing.

Conclusion

NatureScope is feasible both technically and financially. An RK3576-based MVP with 8MP/13MP camera, 2.8" TFT, and GPS can be developed within a \$40 BOM target. The AI pipeline is viable for real-time ID (~250 ms latency). Dataset challenges remain the biggest bottleneck, requiring phased expansion. With iterative improvements, NatureScope can evolve into a global educational and conservation tool.

Summary Table

Aspect	Key Points
Processor	RK3576 (6 TOPS NPU), scalable to RK3588
Camera	Sony IMX219 (8MP) or OV13855 (13MP)
Display	2.8" TFT LCD (480x480)
Battery	3000mAh Li-Po (~4–6 hrs use)
Estimated Cost	\$35–40 BOM, ~\$70–93 prototype
Inference Speed	≤250 ms per frame